



Animation System

Basics of Animation Technology



Humans have been trying to represent object in motion



Stone age mural in the Cave of Altamira

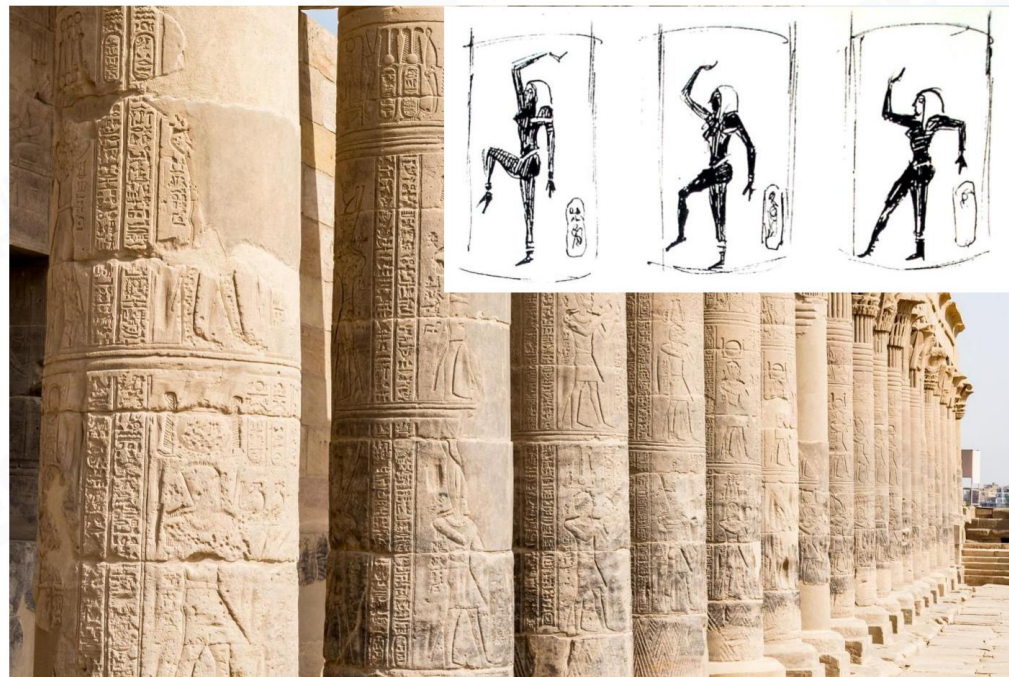


Painted Pottery of
Majiayao Culture

馬家窑文化



Ancient Greek
Pottery



Temple of Isis

Isis: 마법과 모성의 상징으로 오시리스의 아내이자 호루스의 어머니인 이집트 신화의 여신



Humans have been trying to represent object in motion

- The persistence of vision



Discoverer of "*the persistence of vision*"
Peter Mark
Roget(1779-1869)

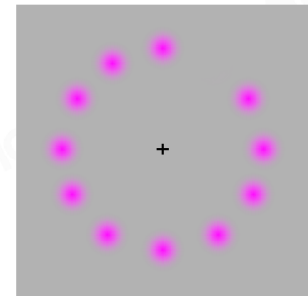


Thaumatrope

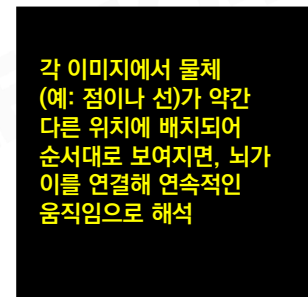
- Illusory motion



Discoverer of "*phi phenomenon*"
Max Wertheimer
(1880-1943)



Phi phenomenon

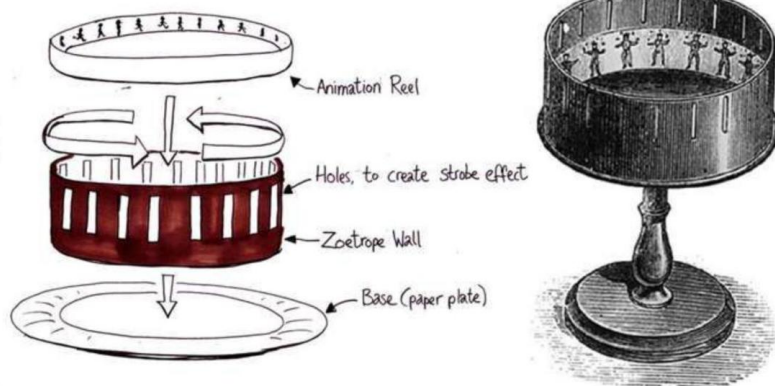


Beta movement

정지된 자극(예: 빛이나 이미지)이 빠르게 번갈아 제시될 때, 그 사이를 움직이는 듯한 착시적 운동(apparent motion)을 느끼는 광학적 착시

인간의 눈과 뇌가 자극이 사라진 후에도 이미지를 일시적으로(약 1/16~1/25초) 유지하는 광학적 착시 현상

Humans have been trying to represent object in motion



Zoetrope

광학 장난감(optical toy)으로, 정지된 이미지 시퀀스를 빠르게 회전시켜 착시적 움직임(애니메이션)을 만드는 장치

1834년 영국 수학자 윌리엄 조지 호너(William George Horner)가 발명했으며, 그리스어 'zōē'(생명) + 'tropos'(회전)에서 유래한 이름



Praxinoscope

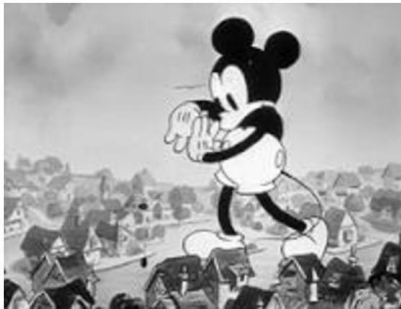


Flip Book

영화와
애니메이션의
모태



Animation Techniques in Film



Gulliver Mickey(1934)



Westworld(1973)



Futureworld(1976)



Jurassic Park(1993)



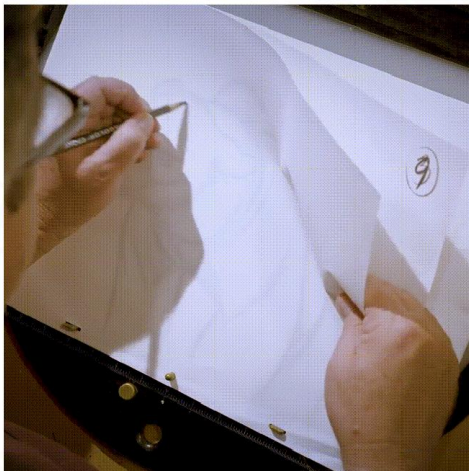
Avatar(2009)



Zafari(2018)



Animation Techniques in Film



Hand Draw Animation



Cel Animation



Computer Animation



Animation Techniques in Game

멀티뷰 스프라이트 애니메이션(multi-view sprite animation)은 2D 게임 개발에서 캐릭터나 오브젝트를 여러 시점(뷰, 예: 정면, 측면, 후면, 사각 등)으로 본 스프라이트 시퀀스를 제작하고 재생하는 애니메이션 기법



Sprite Animation in
Pac-Man(1980)



Sprite Animation Based on
Rotoscoping in
Prince of Persia(1989)



Multi-view Sprite Animation in
Doom(1993)

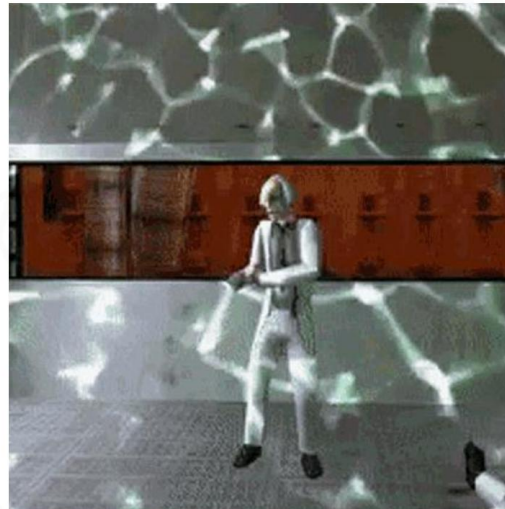
실사 영상을 프레임별로 추적(tracing)하여 자연스러운 움직임을 재현한
2D 스프라이트 시퀀스 애니메이션



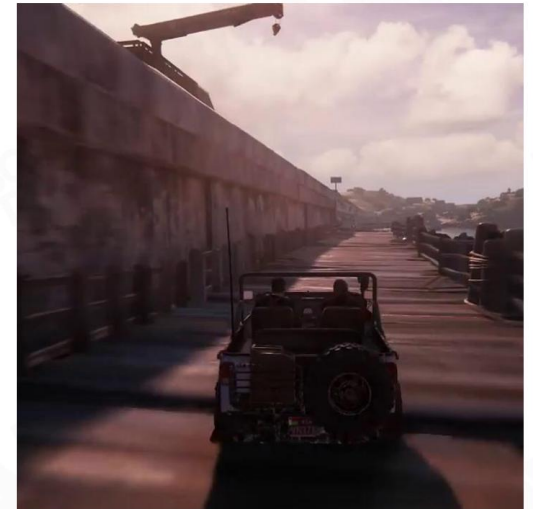
Animation Techniques in Game



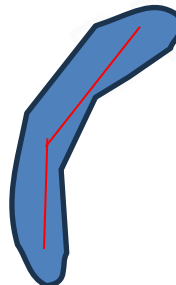
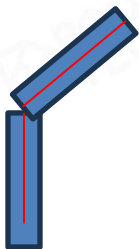
Rigid Hierarchy Animation in
Resident Evil(1996)



Soft Skinned Animation in
Half-life(1998)



Physics Animation in
Uncharted 4(2016)





Outline of Animation System

01.

Basics of Animation Technology

- 2D Animation
- 3D Animation
- Skinned Animation Implementation
- Animation Compression
- Animation DCC

02.

Advanced Animation Technology

- Animation Blend
- Inverse Kinematics
- Animation Pipeline
- Animation Graph
- Facial Animation
- Retargeting



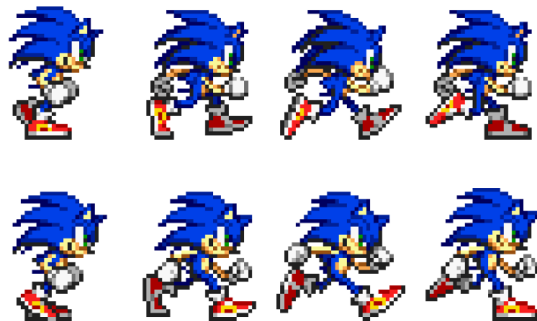
2D Animation Techniques in Games



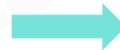
2D Animation - Sprite animation

The electronic equivalent to cel animation

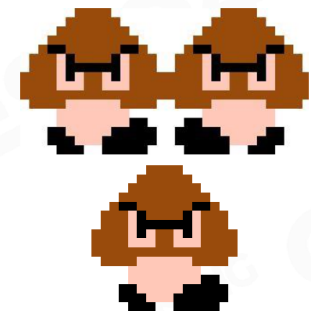
- A sprite is a small bitmap that can be overlaid on top of a background image without disrupting it
- The sequence of frames was designed so that it animates smoothly even when it is repeated indefinitely



Sprite Sequence



Loop



The mushroom in
Super Mario Bros.

The Sprite-like animation technique in pseudo-3D game



A sprite sheet in *Doom*



Multiple views of the sprite animation

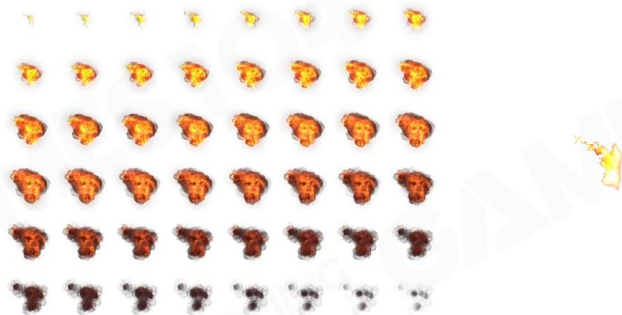


Sprite animation in *Doom*

Sprite Animation in Modern Game

Application

- 2D character
 - Sprite on 2D background image
 - Sprite on top of 3D rendered environment
- Game effect
 - Sprite sheet texture for particles



Sprite animation for particle effect



2D character with 3D environment
in *Octopath Traveler*

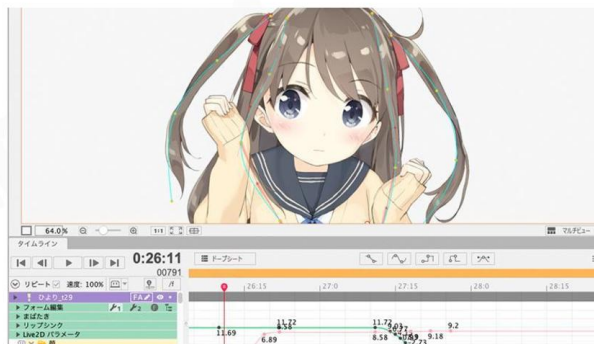
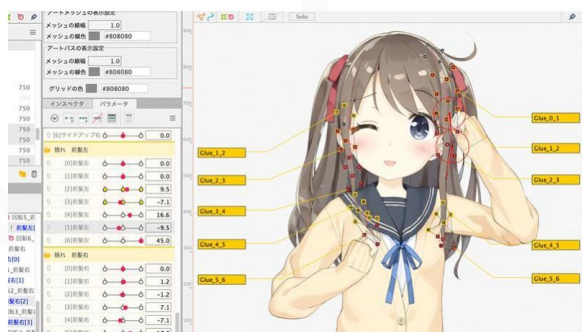


2D game *To the Moon*

Live2D

A technology to generate 2D animation without 3D model

2D 일러스트를 3D처럼 동적으로 변형(deform)하여 애니메이션을 생성하는 이는 전통적인 2D 애니메이션(프레임별 수작업 그리기)과 달리, 메시(mesh)와 파라미터(parameter)를 활용해 실시간 변형을 가능하게 하며, Live2D Cubism 같은 소프트웨어로 구현



 Live2D

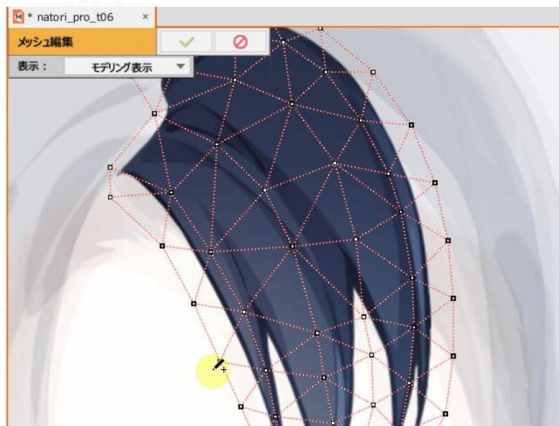


Live2D

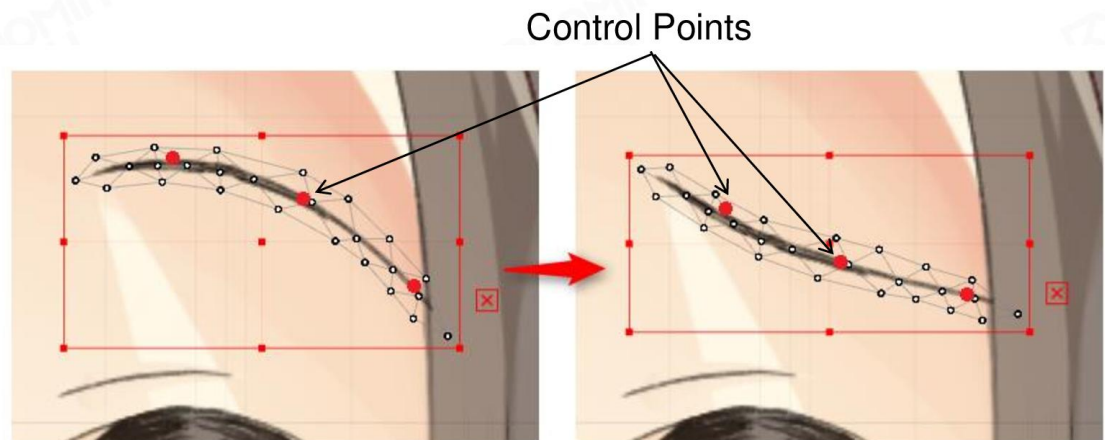
Make a Live2D animation

Transform image by using control points for parts

- "ArtMesh" could be automatically generated for each parts, which defined by vertices, edges and polygons
- Control points could be used to help transforming "ArtMesh"



Adjust automatically generated "ArtMesh"



Use control point to change "ArtMesh"

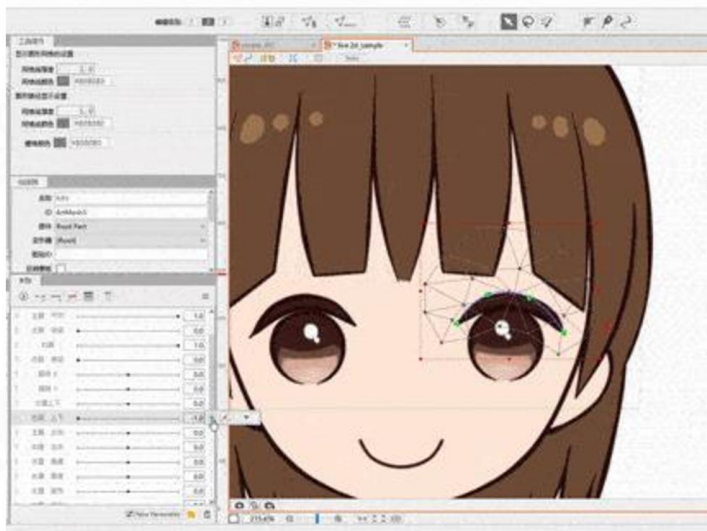


Live2D

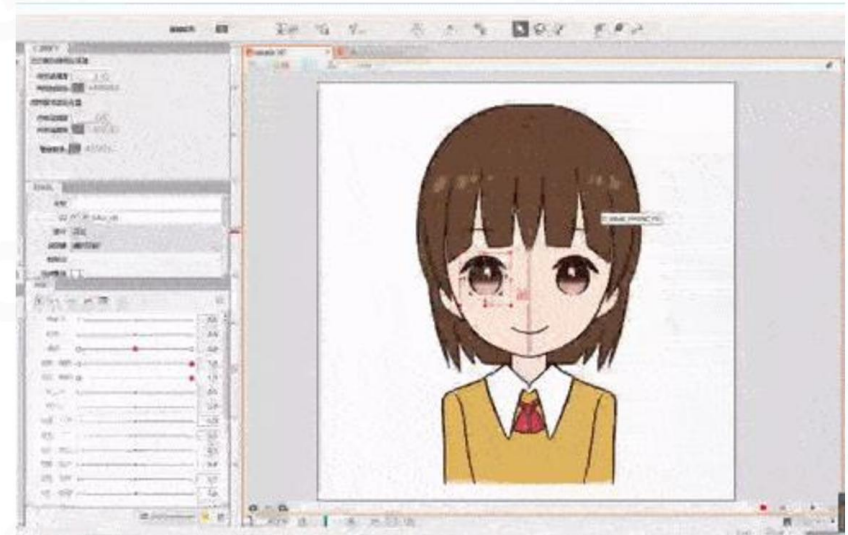
Make a Live2D animation

Set animation "key frame"

- Set "key frame" to help animation interpolation



Set "key frame" of animation



Final animation



3D Animation Techniques in Games



DoF(Degrees of Freedom)

- refers to the number of independent variables or parameters of a system



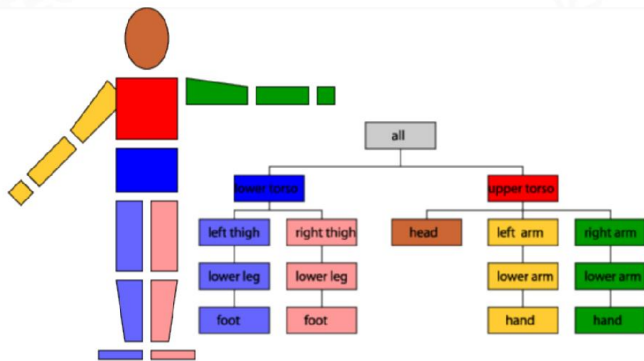


Rigid Hierarchical Animation

- The earliest approach to 3D character animation
- A character is modeled as a collection of rigid pieces
- The rigid pieces are constrained to one another in a hierarchical fashion



Chinese leather-silhouette show



General rigid hierarchy



Rigid hierarchical animation
in *Resident Evil*(1996)

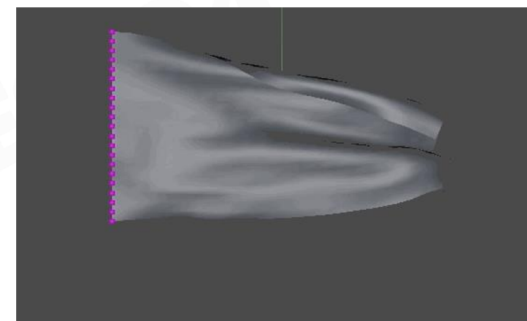


Cracking at the joints

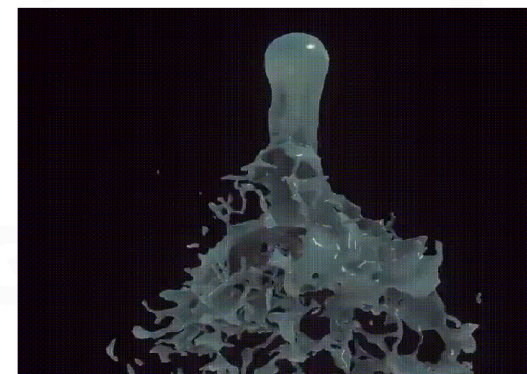


Per-vertex Animation

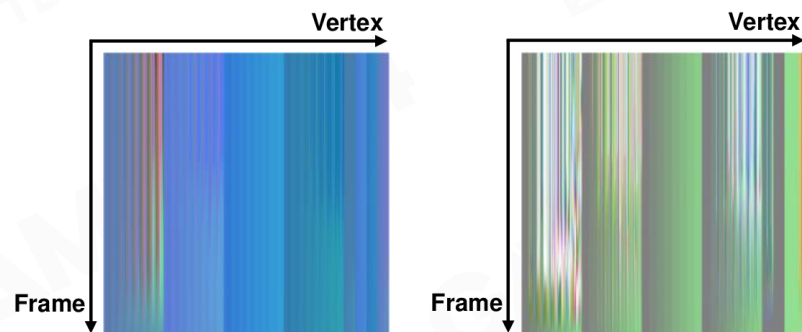
- Most flexible(3 DoFs per vertex)
- Mostly implemented by Vertex Animation Texture(VAT)
- Suitable for complex morphing
- Need massive data



Cloth Vertex Animation



Fluid Vertex Animation



An example of VAT: Translation Texture(left) and Rotation Texture(right), e.g. the texel at (5,8) stores the data at the 9th frame for vertex with index 5



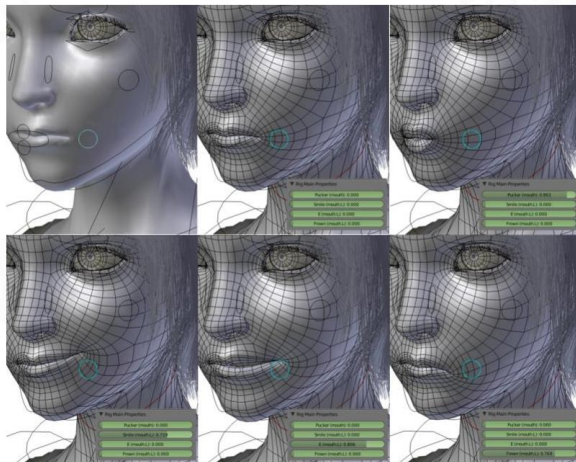
Morph Target Animation

- A variation on Per-vertex Animation
 - Use key frames with LERP instead of sequence frames(e.g. 30 frames per second)
- Suitable for facial expression

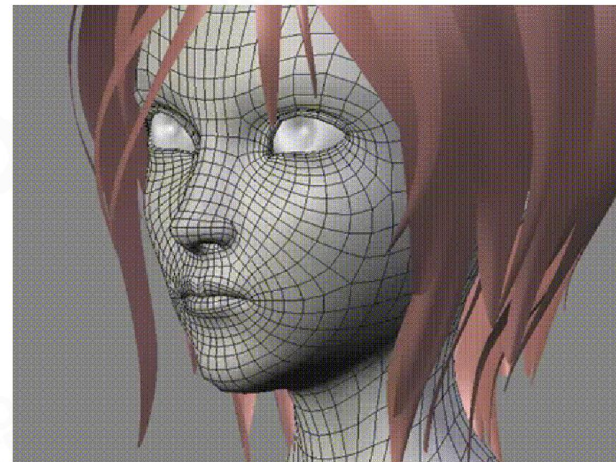
+ Key Frame LERP

Per-Vertex Animation

Morph Target Animation



Key Poses



Play with LERP

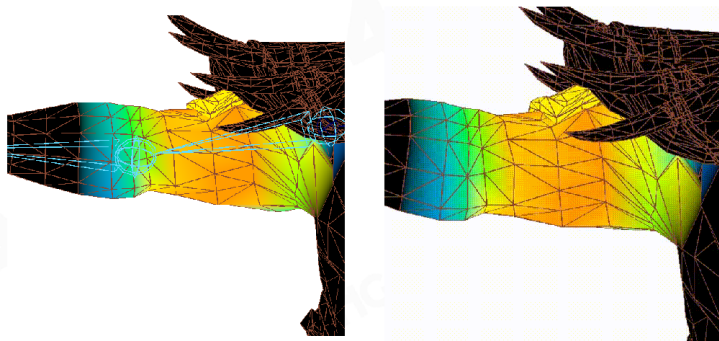


3D Skinned Animation

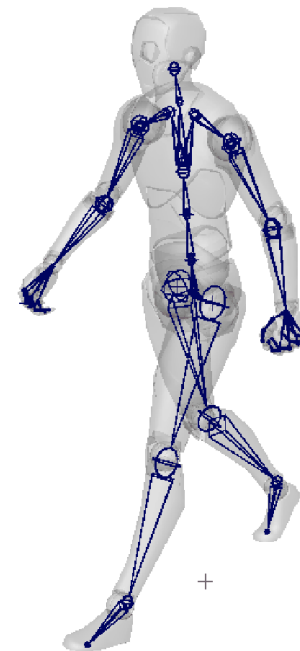
- Mesh(or skin) is bound to the joints of the skeleton
- Each vertex can be weighted to multiple joints

Advantages

- Need less data than per-vertex animation
- Mesh can be animated in a natural way (like human “skin”)



Animation is continuous and smooth at joints



A Skinned Animation of Walking



2D Skinned Animation

Derived from 3D skinned animation

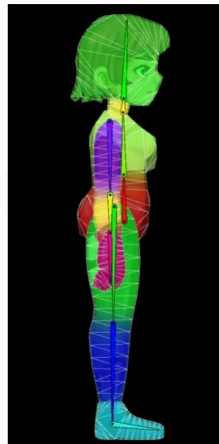
- Break up character into various body parts
- Create body part meshes and piece them together
- Rigging, skinning and animation



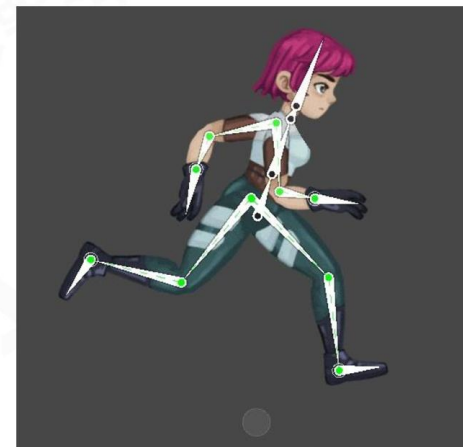
Body parts



Combine parts



Rigging and Skinning

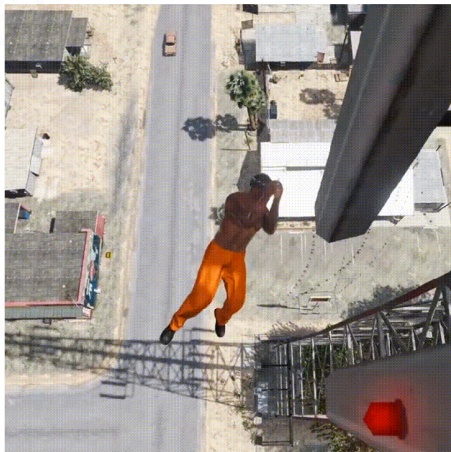


Animation



Physics-based Animation

- Ragdoll
- Cloth and Fluid simulation
- Inverse Kinematics(IK)



Ragdoll in *GTA 5*



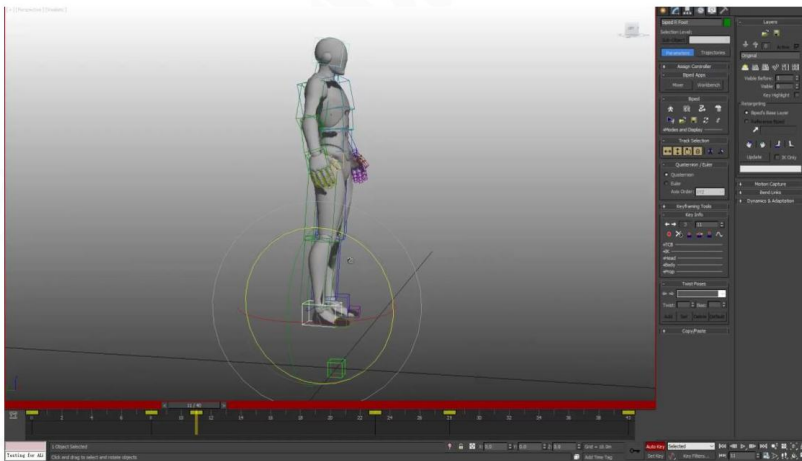
Cloth in Valley of The Ancient from UE5



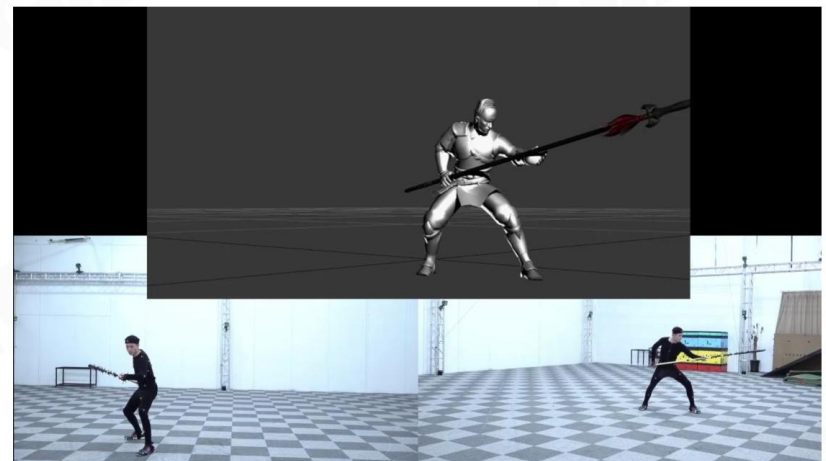
Animation with IK in *Uncharted 4*

Animation Content Creation

- Digital Content Creator + Animator
- Motion Capture



Animation from DCC



Animation from Motion Capture

Digital Content Creation는 디지털 콘텐츠를 생성·편집하는 소프트웨어 도구(예: Blender, Maya, Houdini 등)를 의미하는 특별한 용어



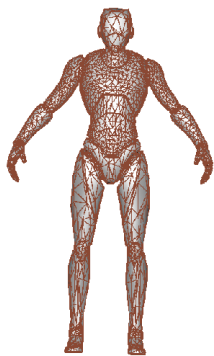
Skinned Animation Implementation



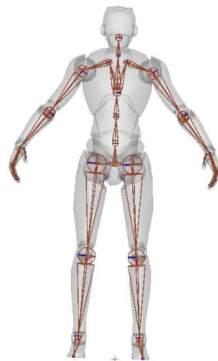
How to Animate a Mesh

1. Create mesh for a binding pose
2. Create a binding skeleton for the mesh
3. "Paint" per-vertices skinning weights to related skeleton
4. Animate skeleton to desired pose
5. Animate skinned vertices by skeleton and skinning weights

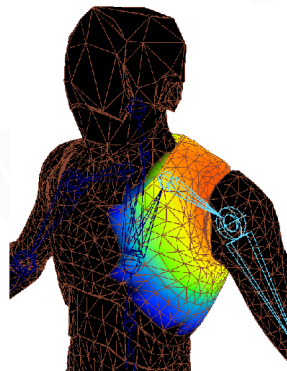
**Sounds Easy
but Not Simple**



Mesh



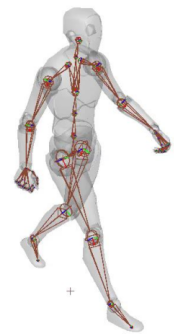
Skeleton



Skinning/Rig



Pose

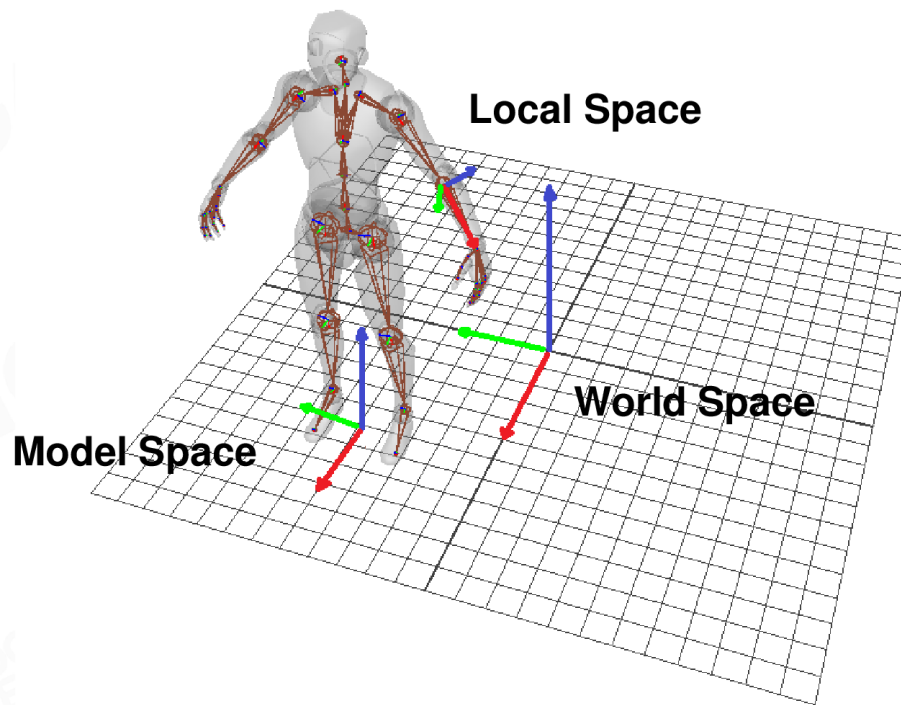


Animation



Different Spaces

- Local space , Model space and World space

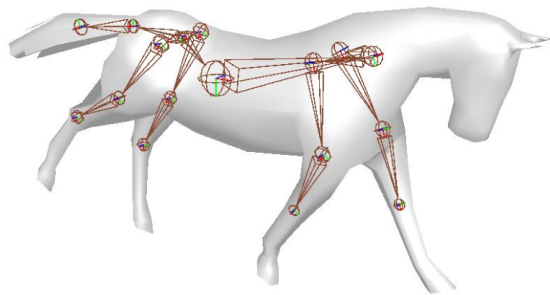




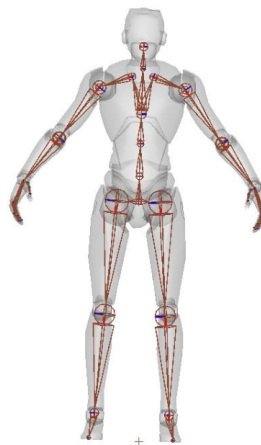
Skeleton for Creatures

Comprised of a hierarchy of rigid pieces known as joints

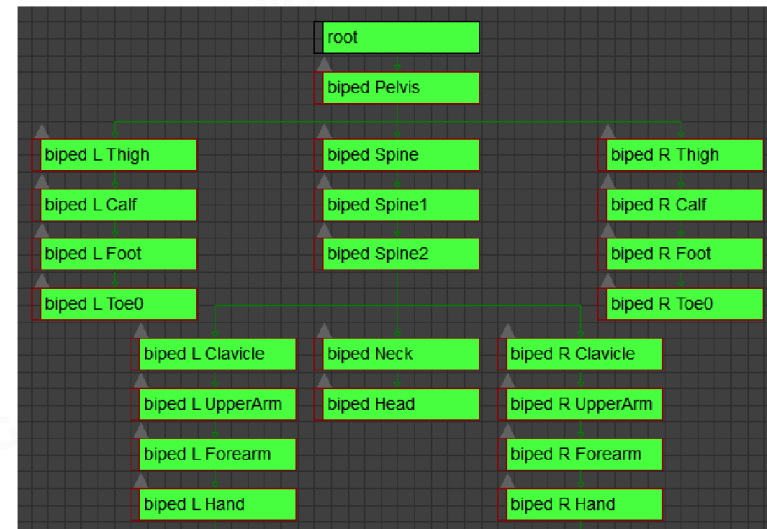
- One joint is selected as the root
- Every joint has a parent joint except the root



Non-humanoid Skeleton



Humanoid Skeleton

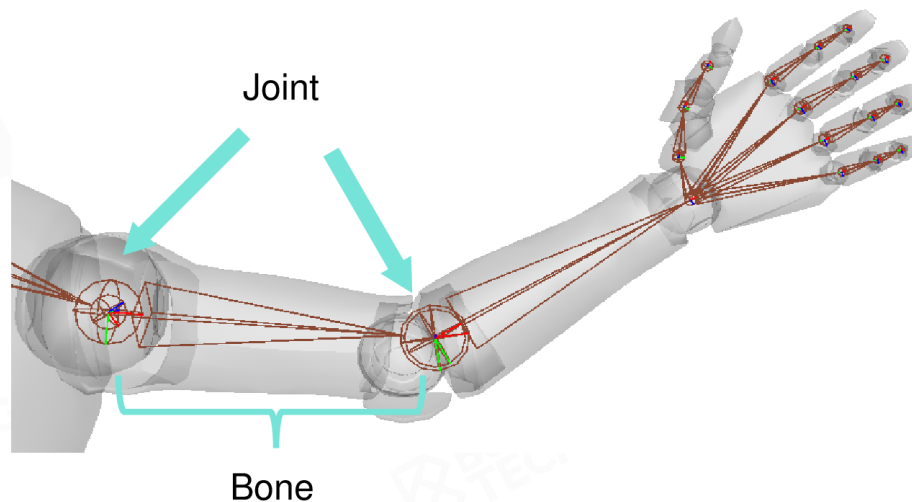


Skeleton Hierarchy



Joint vs. Bone

- The joints are the objects directly manipulated by the animator to control motion
- The bones are the empty space between the joints

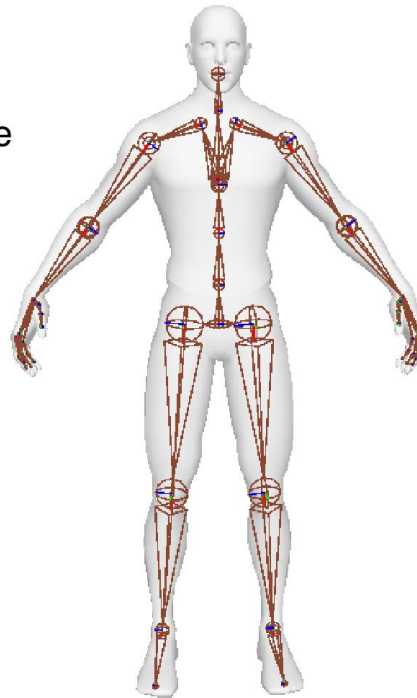




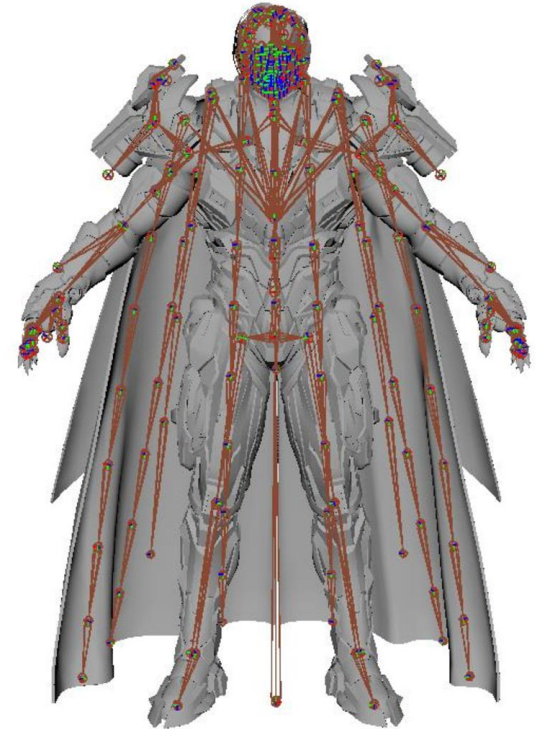
Humanoid Skeleton in Real Game

Number of joints in a humanoid skeleton

- Normal : 50~100 joints
- May more than 300+ joints include facial joints and gameplay joints



58 joints



320 joints



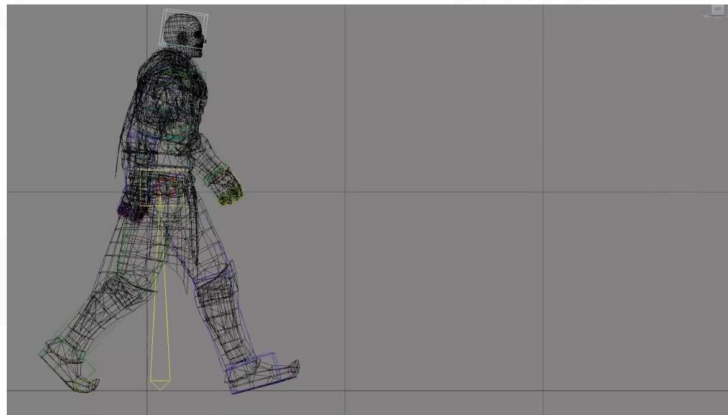
Where to Start the Skeleton – Root Joint

Root joint

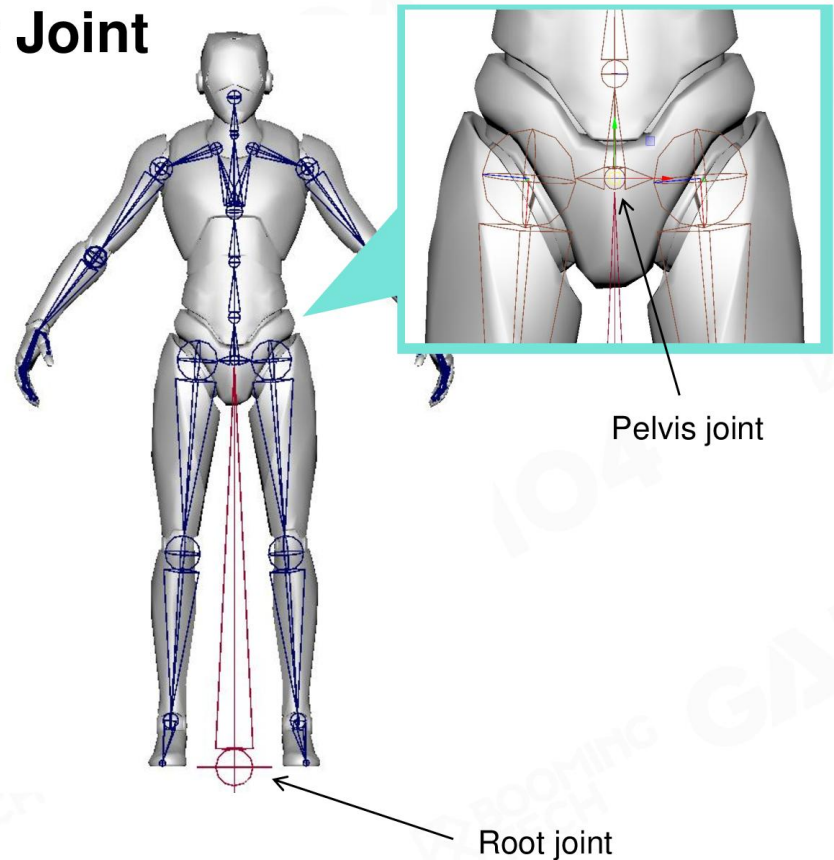
- The center of the feet
- Convenient to touch the ground

Pelvis joint

- The first child joint of the root joint
- Human upper and lower body separation



Root joint move





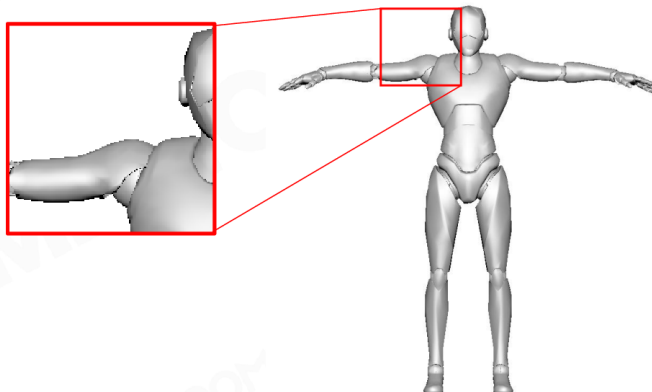
Bind Pose – T-pose vs. A-pose

The pose of the 3D mesh prior to being bound to the skeleton

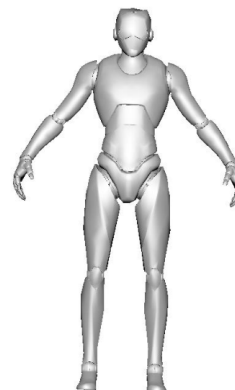
- Keep the limbs away from the body and each other, making the process of binding the vertices to the joints easier
- Usually close to natural pose

T-pose vs A-pose

- Shoulders in A-pose are more relaxed
- Easy to defarmating in A-pose



T-pose



A-pose

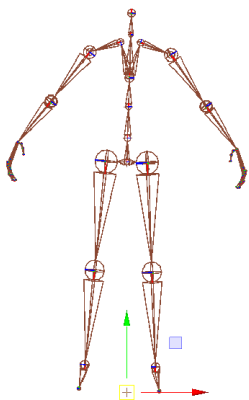


Skeleton Pose

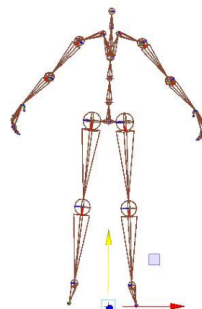
Skeleton Pose : A skeleton is posed by transform its joints from the bind pose

Joint Pose (9 DoFs)

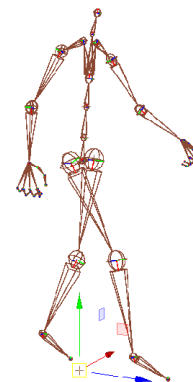
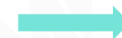
- Orientation (3 DoFs)
- Position (3 DoFs)
- **Scale (3 DoFs)**



Bind Pose



Posed by Transform Joints



The Pose of Walking

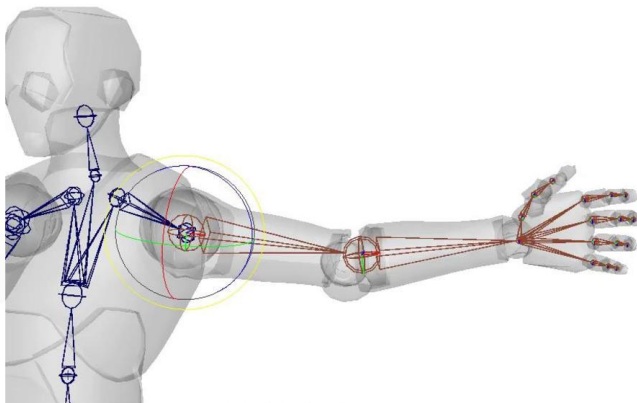


Joint Pose

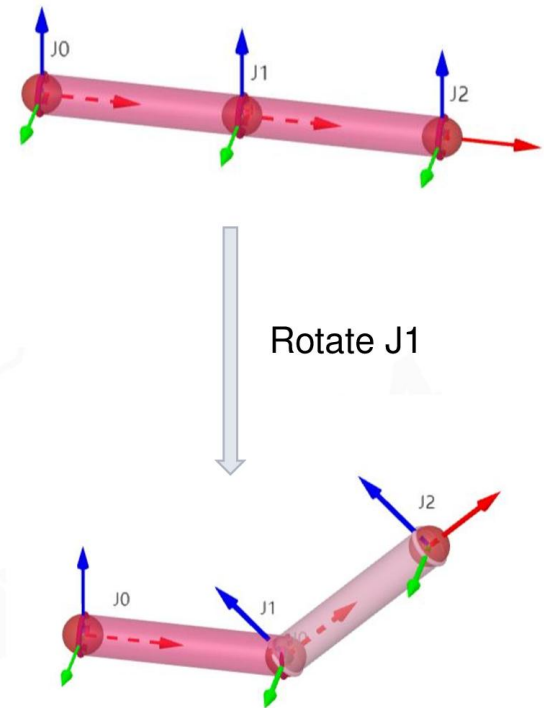


Joint Pose - Orientation

- Rotation -> Change the Orientation of joints
- Most skeleton poses change orientations of joints only



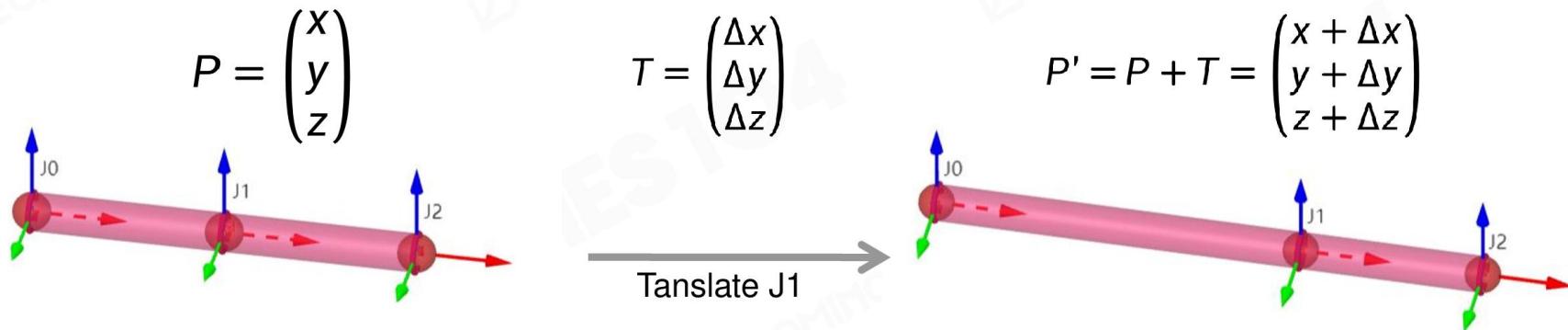
Rotate the forearm





Joint Pose - Position

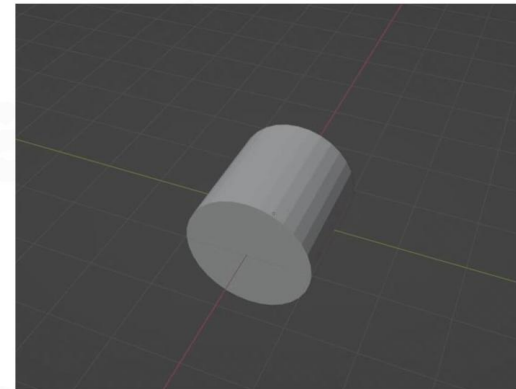
- Translate -> change position
- Translate point P to point P' by vector T





Joint Pose - Scale

- Scale -> change the size of the model
- Uniform vs. Non-uniform Scale

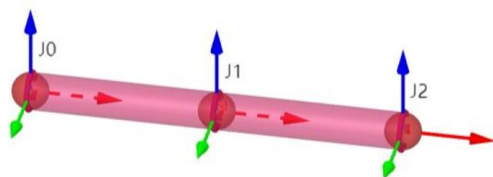


Non-uniform Scale

$$V = \begin{pmatrix} x \\ y \\ z \end{pmatrix}$$

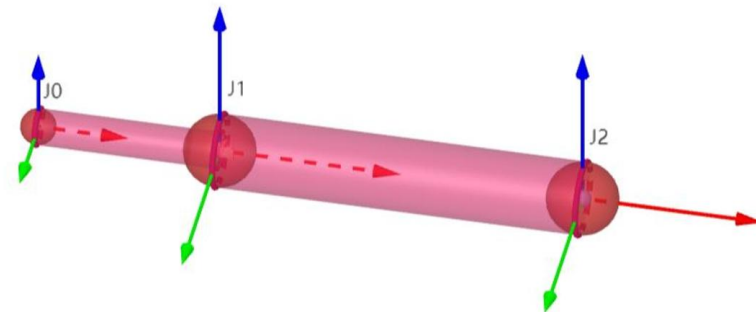
$$S = \begin{pmatrix} a \\ b \\ c \end{pmatrix}$$

$$P_1' = \begin{pmatrix} ax \\ by \\ cz \end{pmatrix}$$



(Uniform Scale: $a = b = c$)

Scale J1





Joint Pose - Affine Matrix

Rotation

$$Q = \begin{pmatrix} a \\ b \\ c \\ d \end{pmatrix} \Rightarrow R = \begin{bmatrix} 1 - 2c^2 - 2d^2 & 2bc - 2ad & 2ac + 2bd \\ 2bc + 2ad & 1 - 2b^2 - 2d^2 & 2cd - 2ab \\ 2bd - 2ac & 2ab + 2cd & 1 - 2b^2 - 2c^2 \end{bmatrix} \Rightarrow$$

$$R_{HM} = \begin{bmatrix} 1 - 2c^2 - 2d^2 & 2bc - 2ad & 2ac + 2bd & 0 \\ 2bc + 2ad & 1 - 2b^2 - 2d^2 & 2cd - 2ab & 0 \\ 2bd - 2ac & 2ab + 2cd & 1 - 2b^2 - 2c^2 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

Translation

$$T_v = \begin{pmatrix} \Delta x \\ \Delta y \\ \Delta z \end{pmatrix} \Rightarrow T_{HM} = \begin{bmatrix} 1 & 0 & 0 & \Delta x \\ 0 & 1 & 0 & \Delta y \\ 0 & 0 & 1 & \Delta z \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

Scale

$$S_v = \begin{pmatrix} a_x \\ b_y \\ c_z \end{pmatrix} \Rightarrow S = \begin{bmatrix} a_x & 0 & 0 \\ 0 & b_y & 0 \\ 0 & 0 & c_z \end{bmatrix} \Rightarrow S_{HM} = \begin{bmatrix} a_x & 0 & 0 & 0 \\ 0 & b_y & 0 & 0 \\ 0 & 0 & c_z & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

Affine Matrix

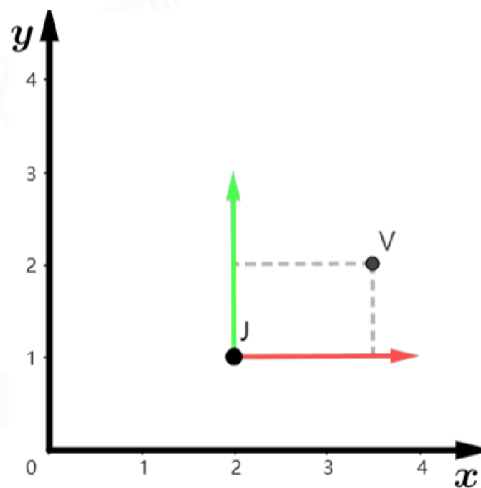
$$M = R_{HM} T_{HM} S_{HM} = \begin{bmatrix} SR & T \\ 0 & 1 \end{bmatrix}$$



Single Joint Skinning

Attach the vertices of a mesh to a posed skeleton

- Each vertex can be bound to one or more joints with a weight parameter
- The vertex position in each bound joint's local space is fixed



Vertex V 's position in joint J 's local space is fixed

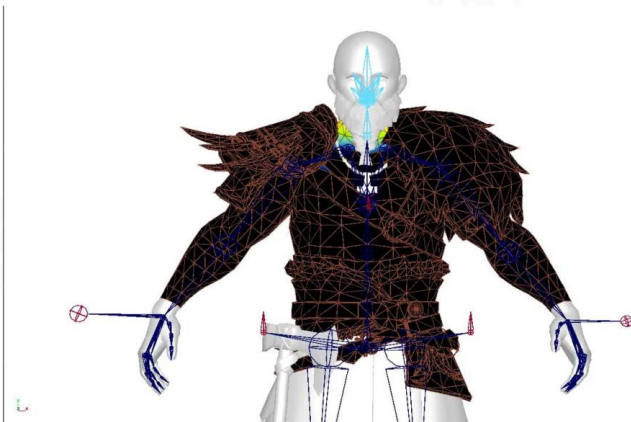


Weighted Skinning with Multi-joints

For a mesh vertex $\mathbf{V}$ which is bound to N joints

- W_i : the **Skinning Weight** of the i -th bound joint

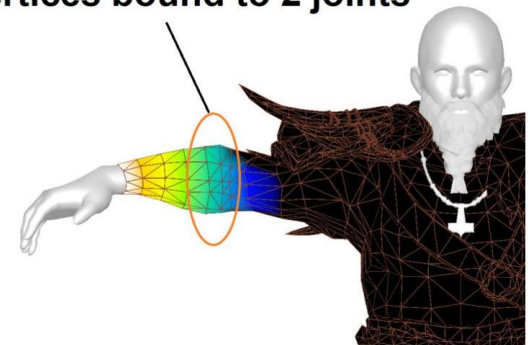
$$\sum_{i=0}^{N-1} W_i = 1$$



Skinning weight of many different joints



Vertices bound to 2 joints

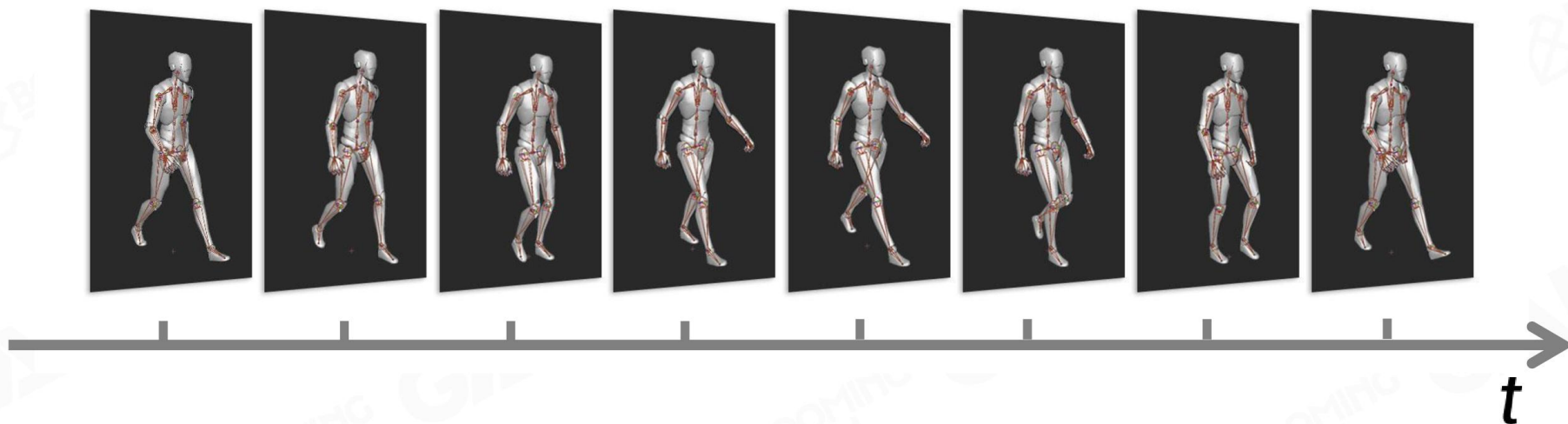


Skinning weight of the upper arm and the fore arm



Clip

A sequence of skeleton poses





Interpolation between Poses

- Animation's timeline is continuous



Pose 1

... ? ...
... Pose 1.5 ...



Pose 2

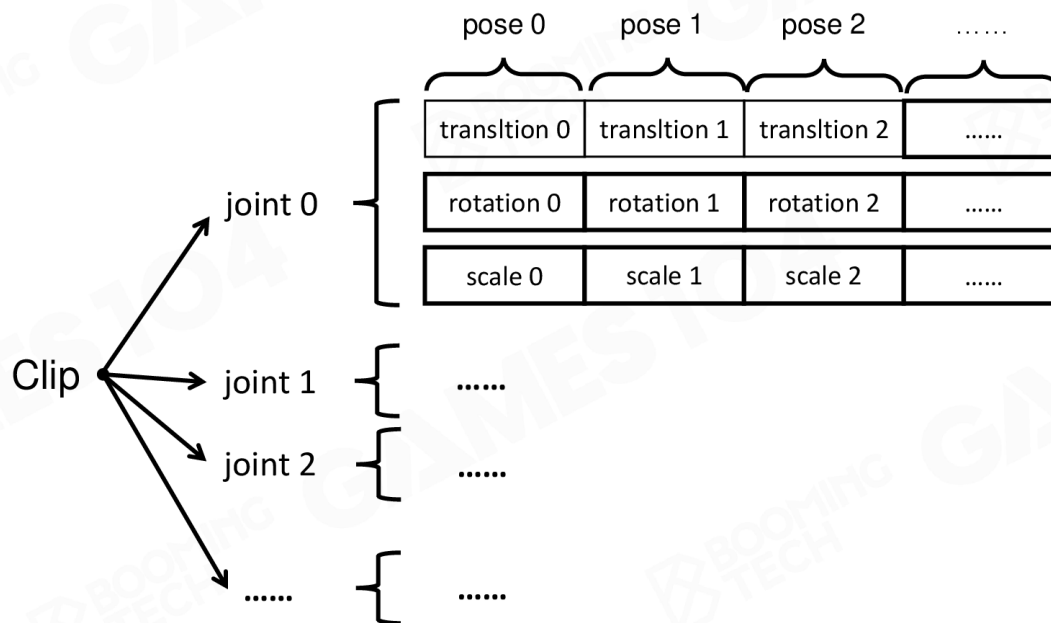


Animation Compression



Animation Clip Storage

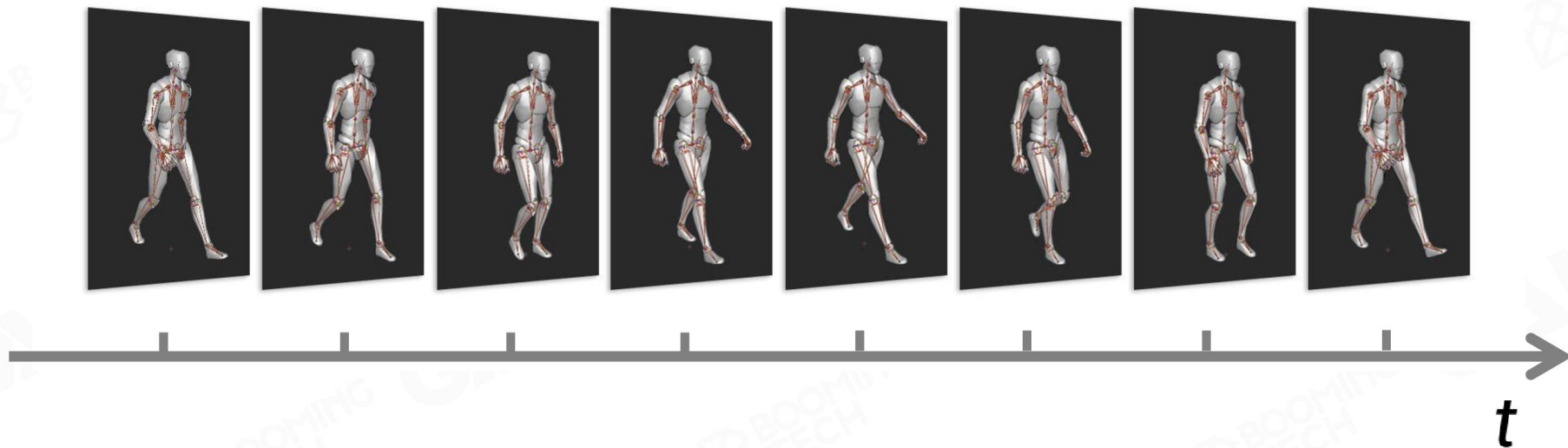
- Animation clip split to separated joint pose sequences
- Joint pose sequence splits to separated translation, rotation and scale tracks





Keyframe

A key frame (or keyframe) in animation and filmmaking is a drawing or shot that defines the starting and ending points of any smooth transition

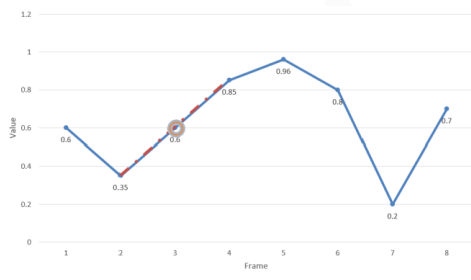




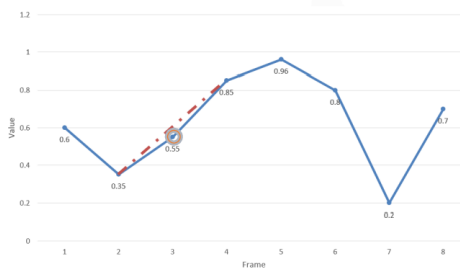
Keyframe Extraction - Linear Keys Reduction

Remove those frames which can be fitted by linear interpolation of adjacent frames

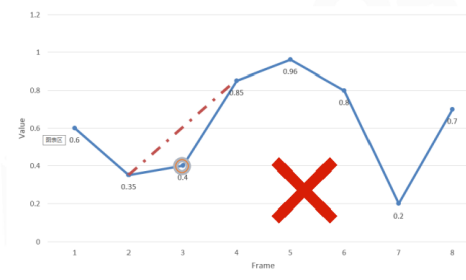
```
KeyFrame = {}  
for i = 1 to n-1 do  
    frame_interp = Lerp(frame[i-1], frame[i+1])  
    error = Diff(frame[i], frame_interp)  
    if isNotAcceptable(error) then  
        KeyFrame.insert(frame[i])  
    end  
end
```



Accurate Case



Acceptable Case



Not Acceptable Case



Animation DCC Process



Animation DCC Process

In general, the Animation DCC(Digital Content Creating) process includes:

- Mesh
- Skeleton binding
- Skinning
- Animation creation
- Exporting



Mesh building

- Blockout Stage: Create a rough outline of the character
- Highpoly Stage: Improve the building precision
- Lowpoly Stage: Devide the surface into meshes
- Texture Stage: Add texture to character

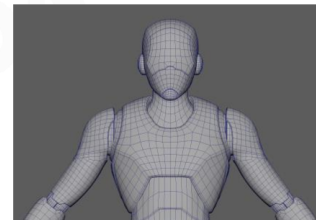
animation related



Blockout Stage



Highpoly Stage



Lowpoly Stage

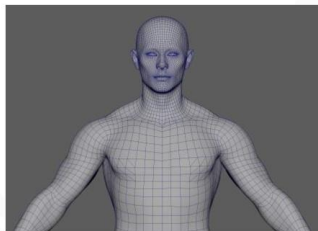


Texture Stage

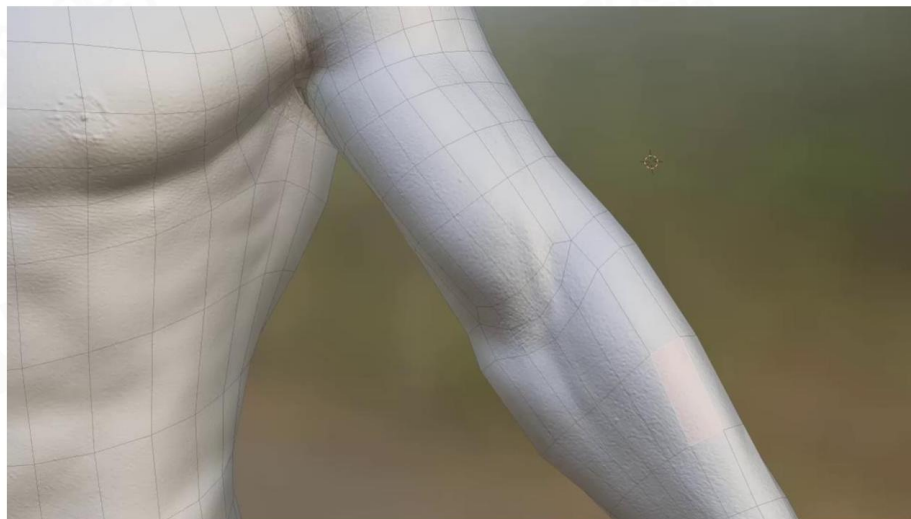


Mesh Adjustment for Animation

- Mesh dividing is vital for animation creating, it defines how the skin curves
- Animation turns to be weird if the meshes are too sparse
- Dense meshes cause a performance overhead



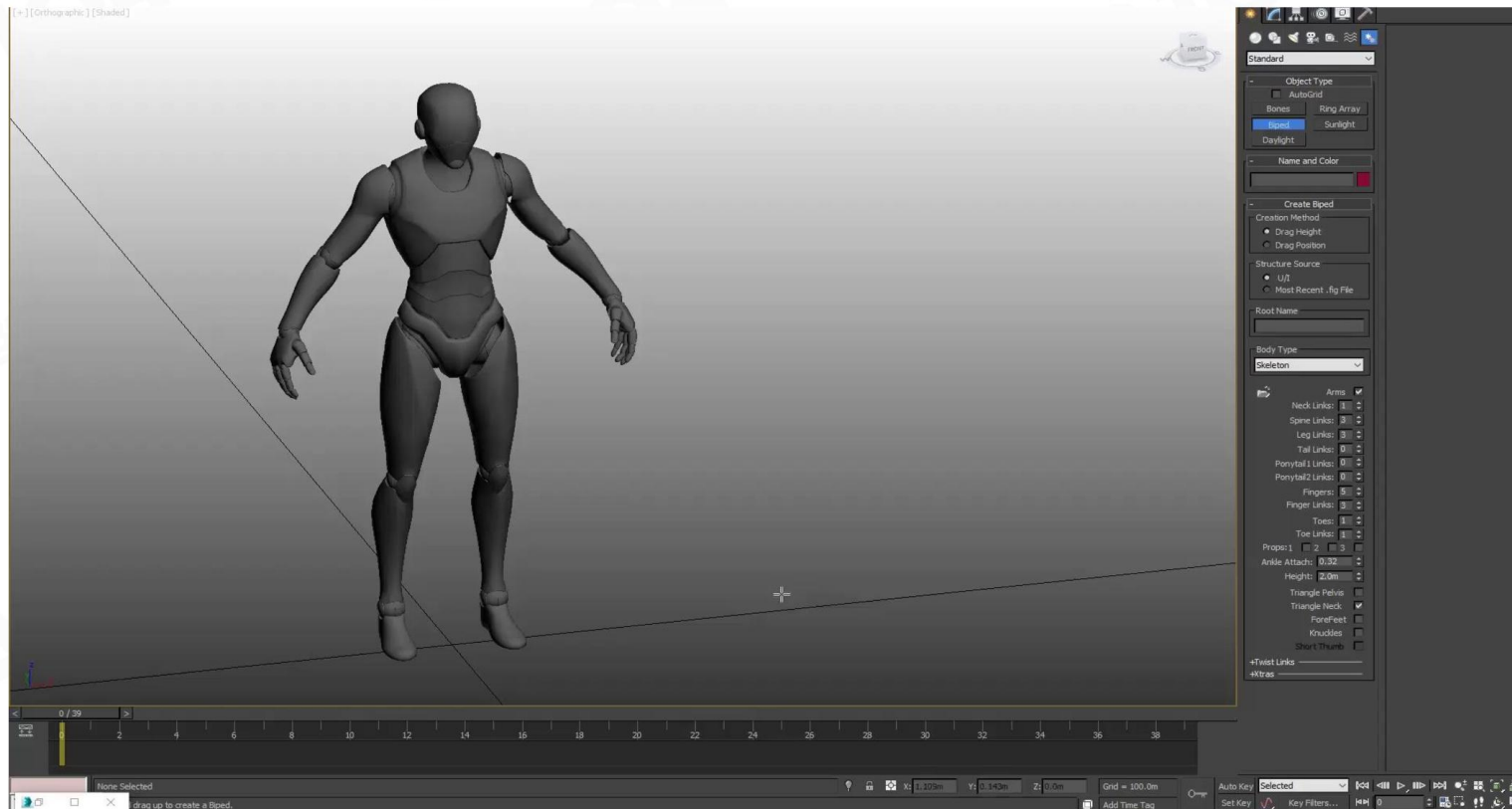
Lowpoly Stage



Mesh Dividing

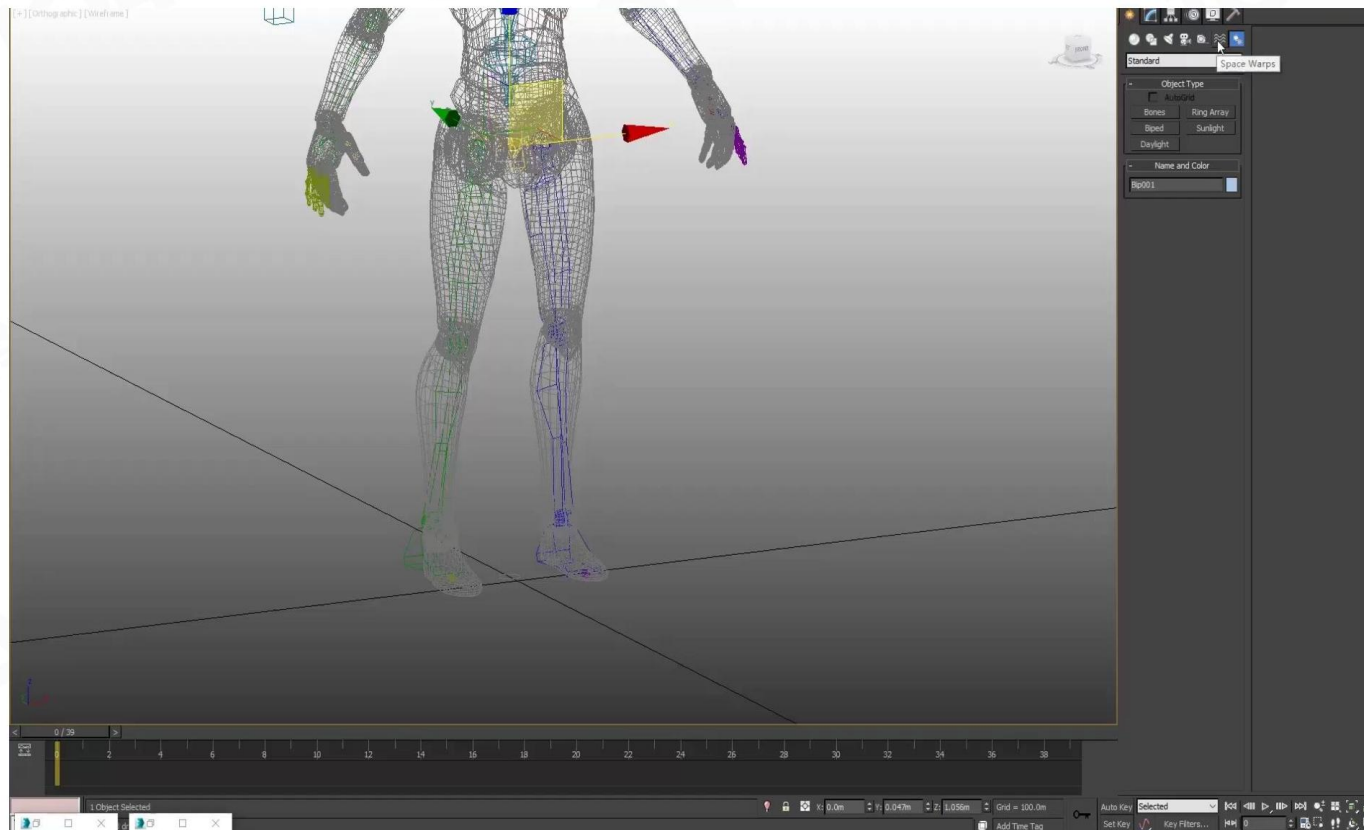


Skeleton Binding



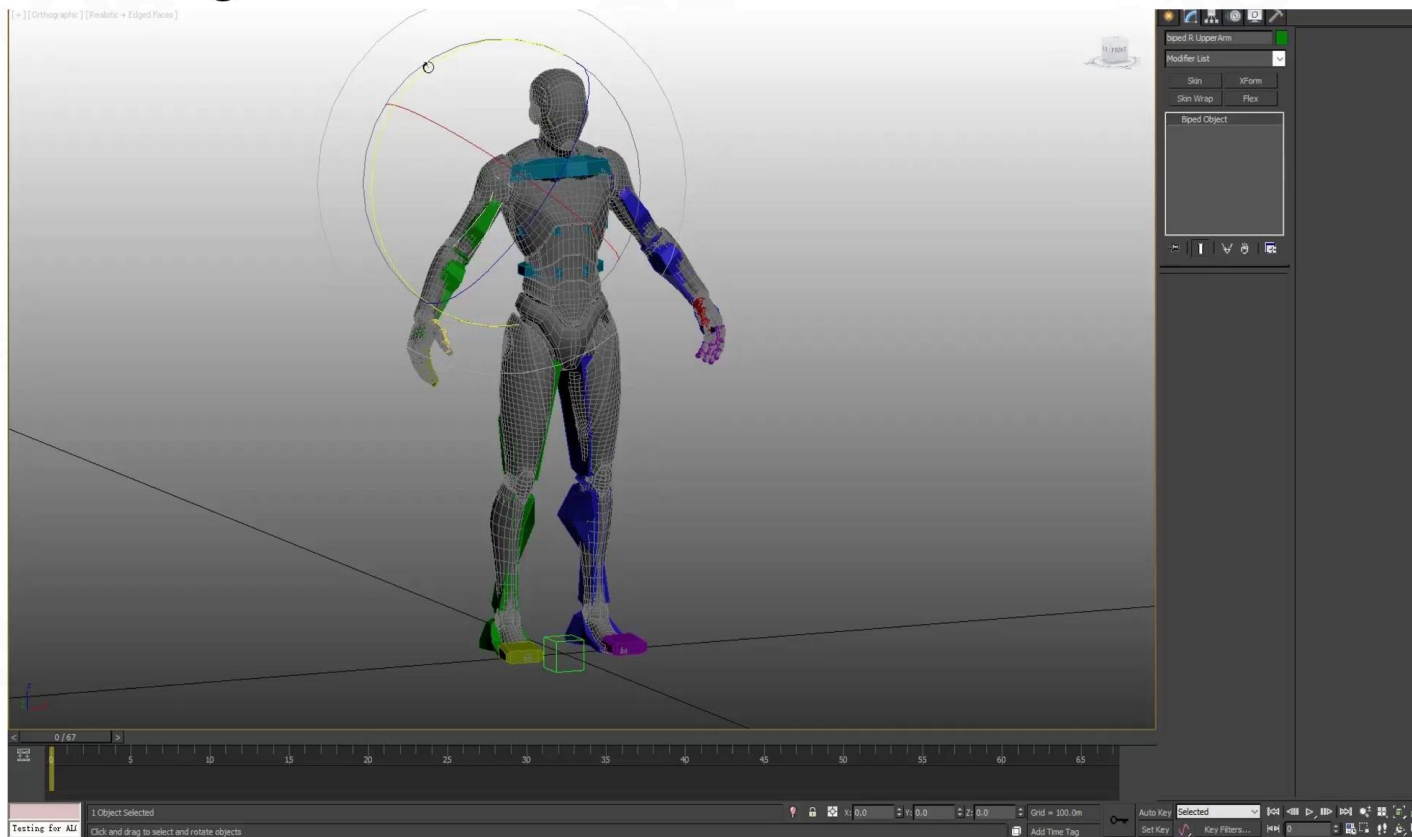


Skeleton Binding – Add Root Joint



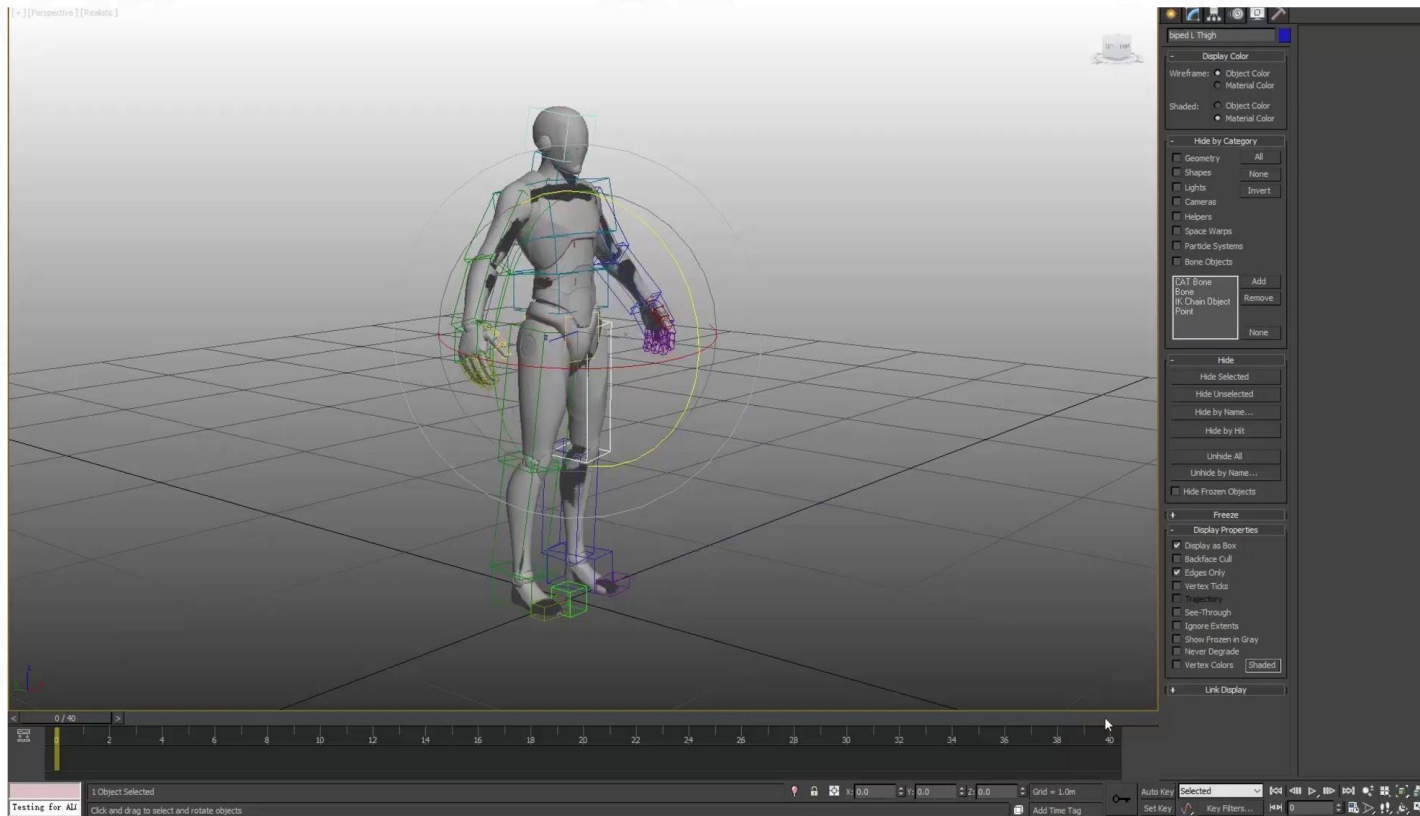


Skinning – Auto Calculation



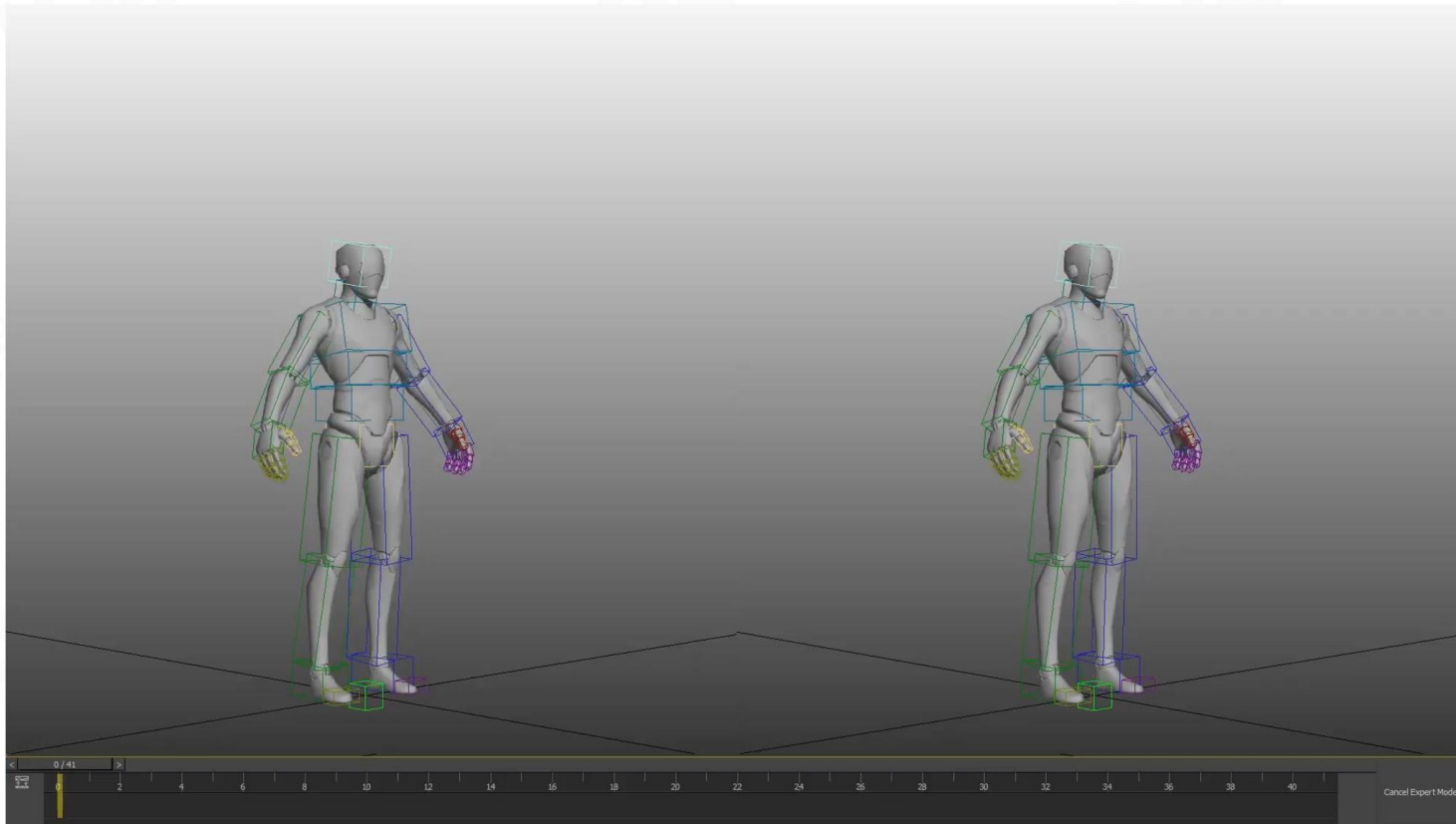


Animation Creation





Exporting



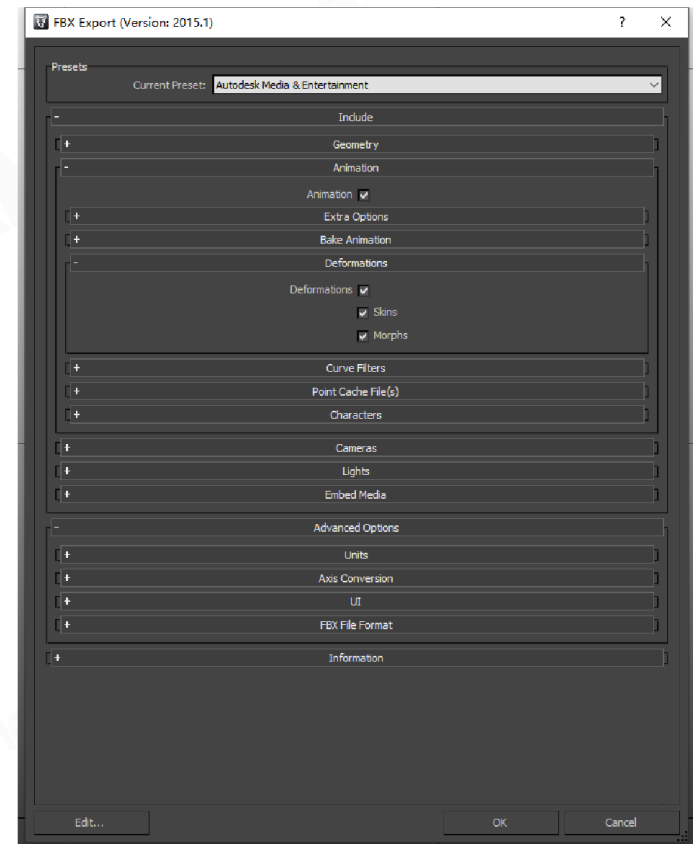


Exporting

Model mesh
skeleton
skinning data
animation clip

} FBX file

FBX: the industry-standard 3D asset exchange file format for games. It is developed by Autodesk as a proprietary format.



Exporting Dialog: Select Asset